



Token. Authentic Shortage

Thank you for listening!

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The background is a dark collage of six images: two tugboats on the water (top left), a wind farm at sea (top right), industrial machinery with pipes (bottom left), a commercial airplane in flight (bottom middle-left), a semi-truck on a highway (bottom middle-right), and an excavator working on a construction site (bottom right).

Importance of Human Aspects in Software Engineering

or

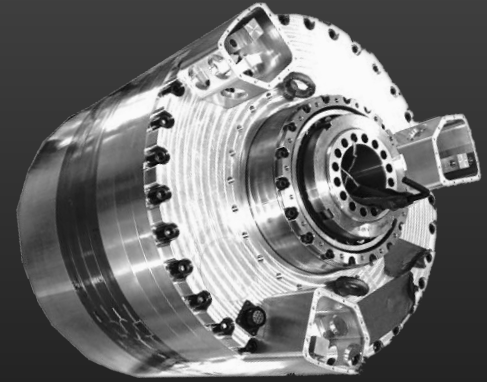
Importance of Behavioral Software Engineering

Per Lenberg

Who is Per Lenberg?

Hagnesia

- Academia
 - Master's degree in Software Engineering
 - Bachelor's degree in Psychology
 - PhD in Software Engineering
- Industry
 - Product Owner at Saab Air Traffic Management
 - CEO Hagnesia



Software Engineering - What & Why

What is Software Engineering?

Software Engineering is the systematic application of engineering principles to define, design, develop, and test software systems.

In short, it's a structured approach to creating reliable software that meets user and business requirements.

Why Software Engineering?

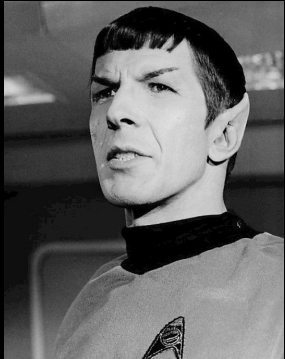
When developing a complex ATM system involving more than 50 software developers, a structured approach is essential.

ATM systems are safety-critical and must comply with regulatory standards, which mandate the use of systematic development processes.

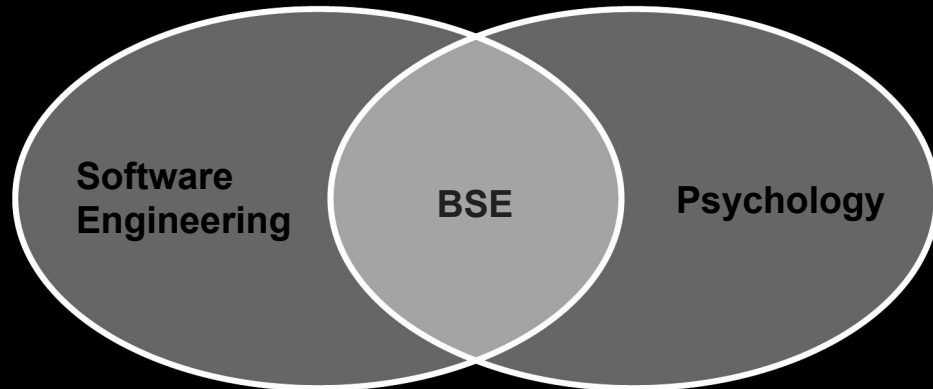
BSE focuses on the human and behavioral aspects of software development processes.

It applies psychological and social principles to address issues affecting software development.

Rational

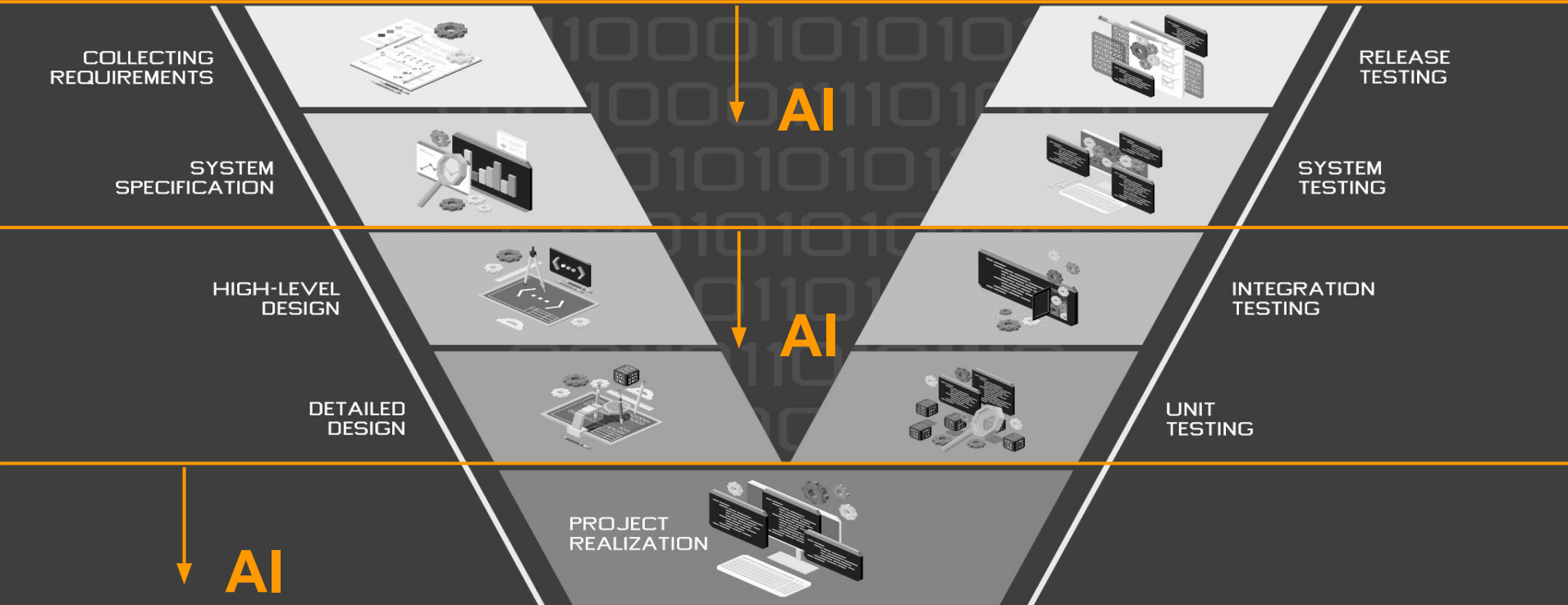


Irrational



What will happen to SE?

V - MODEL SOFTWARE DEVELOPMENT



Semi-executable stack: A diagnostic map of SE

Hagnesia

Before widespread AI use

With AI-mediated work

Code, scripts, tests

Generated/AI-assisted code and tests

Requirements, task descriptions,
checklists, procedures, examples

Prompts, system instructions

Human workflows, handoffs, meetings,
ticket flows, tool chains

Agentic workflows, AI tool use,
human-agent loops

Reviews, approvals, QA gates, audits,
monitoring, escalation

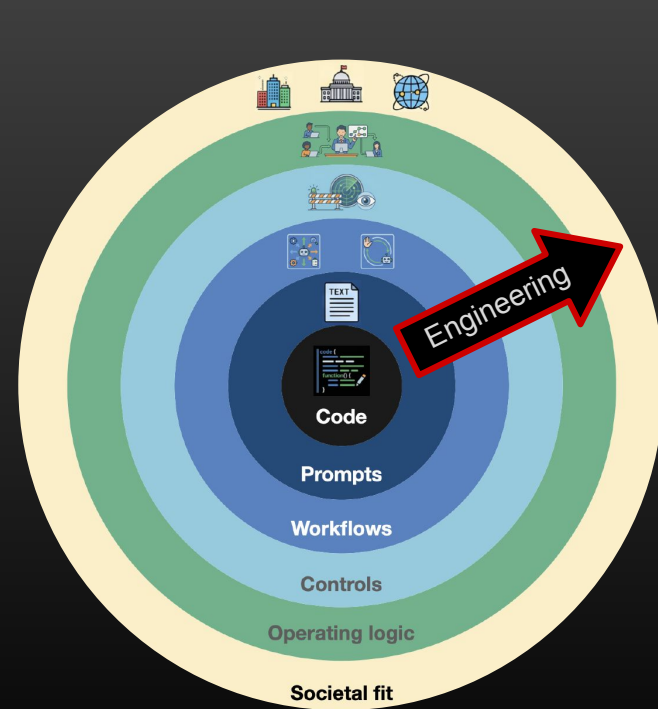
Automated evaluation, confidence
thresholds

Roles, decision rights, delegation,
accountability, "how work really gets
done"

Redesigned roles, AI-assisted decision
preparation, changed accountability

Regulation, customer trust,
professional norms, certification

AI-specific compliance, explainability
demands, AI-mediated decisions



What will happen to software engineers

Hagnesia

Change becomes a persistent condition

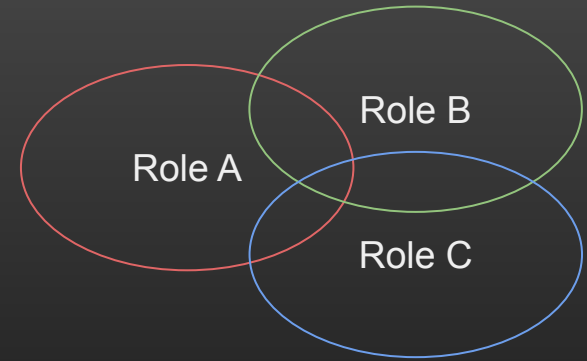
Software Engineers must manage ongoing uncertainty, repeated role adaptation, and continuous learning rather than a one-off change initiative.

Hidden gaps become visible

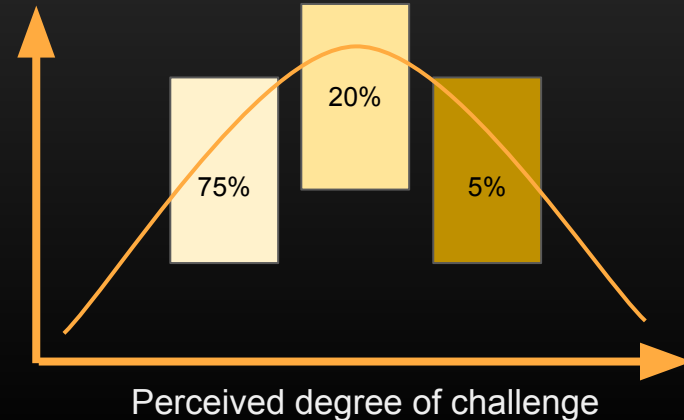
As AI replaces some human interpretation and informal coordination, unclear decisions, responsibilities, and escalation paths become visible.

Less routine work is not automatically better work

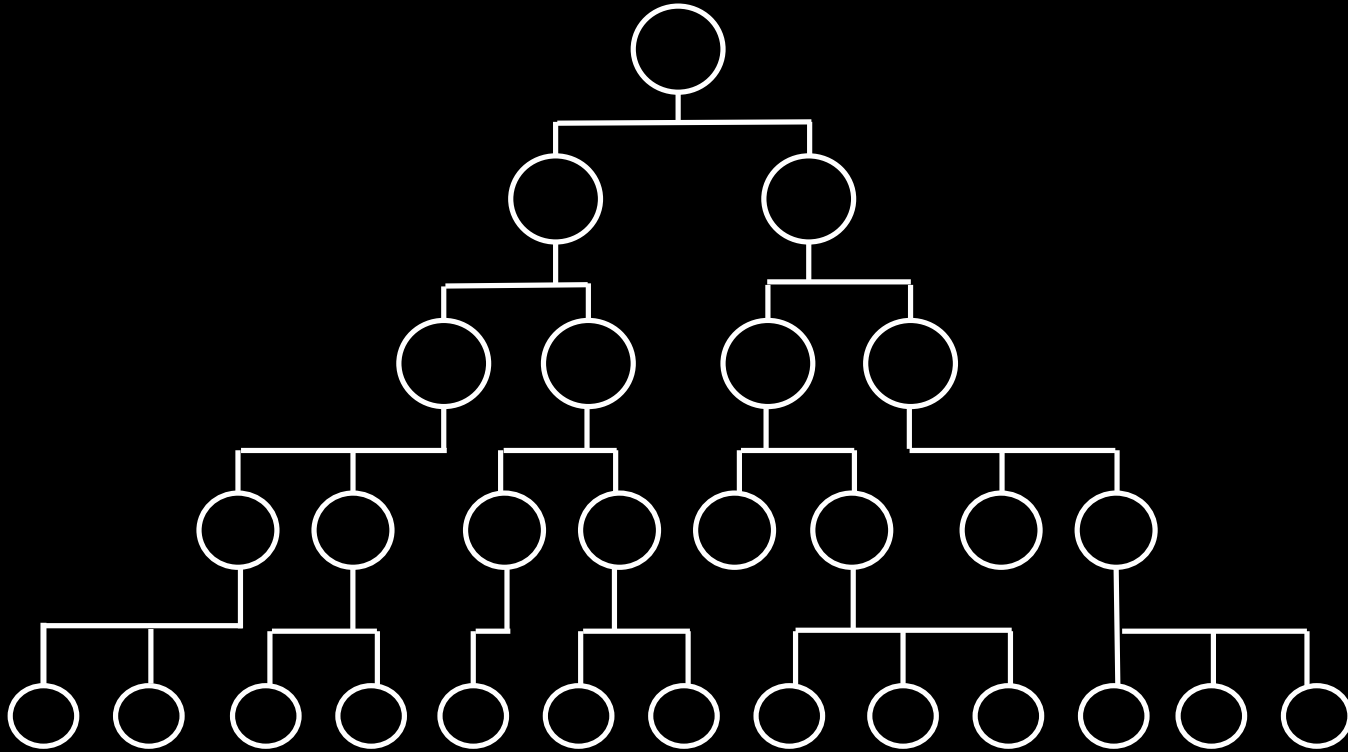
Removing administrative overhead increases the need to design a sustainable balance between cognitive intensity, variation, and recovery.



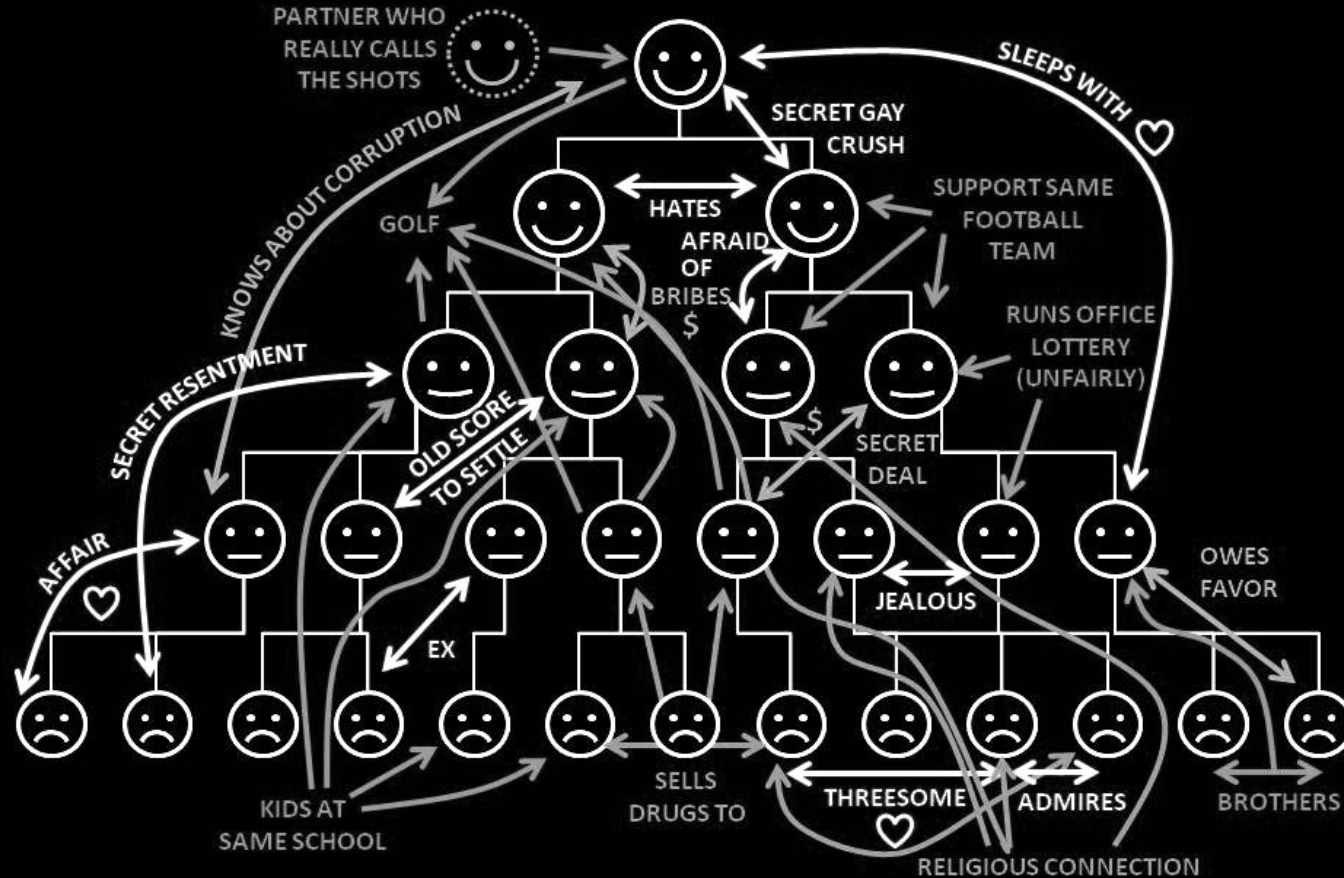
Level of
Motivation



Organizational structure



Organizational structure





Thank you for listening!



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